

BINARY NUMBERS – SHORT REVISION NOTES

Apply Level (Use Knowledge in New Situations)

1. Converting Decimal to Binary (Division Method)

Rule: Divide by 2 repeatedly. Read remainders from BOTTOM to TOP.

Step	Action	Ex.1: 22_{10}	Ex.2: 13_{10}	Ex.3: 42_{10}
1	Divide by 2	$22 \div 2 = 11$ R0	$13 \div 2 = 6$ R1	$42 \div 2 = 21$ R0
2	Continue	$11 \div 2 = 5$ R1	$6 \div 2 = 3$ R0	$21 \div 2 = 10$ R1
3	Continue	$5 \div 2 = 2$ R1	$3 \div 2 = 1$ R1	$10 \div 2 = 5$ R0
4	Continue	$2 \div 2 = 1$ R0	$1 \div 2 = 0$ R1	$5 \div 2 = 2$ R1
5	Continue	$1 \div 2 = 0$ R1	Stop	$2 \div 2 = 1$ R0
6	Read bottom to top	10110_2	1101_2	$1 \div 2 = 0$ R1 $\rightarrow 101010_2$

Quick Application:

Decimal	Remainders	Binary
25_{10}	$25 \div 2 = 12$ R1, $12 \div 2 = 6$ R0, $6 \div 2 = 3$ R0, $3 \div 2 = 1$ R1, $1 \div 2 = 0$ R1	11001_2
30_{10}	$30 \div 2 = 15$ R0, $15 \div 2 = 7$ R1, $7 \div 2 = 3$ R1, $3 \div 2 = 1$ R1, $1 \div 2 = 0$ R1	11110_2
15_{10}	$15 \div 2 = 7$ R1, $7 \div 2 = 3$ R1, $3 \div 2 = 1$ R1, $1 \div 2 = 0$ R1	1111_2
8_{10}	$8 \div 2 = 4$ R0, $4 \div 2 = 2$ R0, $2 \div 2 = 1$ R0, $1 \div 2 = 0$ R1	1000_2
20_{10}	$20 \div 2 = 10$ R0, $10 \div 2 = 5$ R0, $5 \div 2 = 2$ R1, $2 \div 2 = 1$ R0, $1 \div 2 = 0$ R1	10100_2
49_{10}	$49 \div 2 = 24$ R1, $24 \div 2 = 12$ R0, $12 \div 2 = 6$ R0, $6 \div 2 = 3$ R0, $3 \div 2 = 1$ R1, $1 \div 2 = 0$ R1	110001_2

2. Converting Binary to Decimal (Expansion Method)

Rule: Multiply each digit by its place value (power of 2). Add the results.

Step	Ex.1: 1101_2	Ex.2: 10110_2	Ex.3: 1111_2
1. Place values	$2^3, 2^2, 2^1, 2^0$	$2^4, 2^3, 2^2, 2^1, 2^0$	$2^3, 2^2, 2^1, 2^0$
2. Multiply	$1 \times 8, 1 \times 4, 0 \times 2, 1 \times 1$	$1 \times 16, 0 \times 8, 1 \times 4, 1 \times 2, 0 \times 1$	$1 \times 8, 1 \times 4, 1 \times 2, 1 \times 1$
3. Calculate	$8 + 4 + 0 + 1$	$16 + 0 + 4 + 2 + 0$	$8 + 4 + 2 + 1$
4. Add	13_{10}	22_{10}	15_{10}

Quick Application:

Binary	Expansion	Decimal
101_2	$1 \times 4 + 0 \times 2 + 1 \times 1$	5_{10}
10101_2	$1 \times 16 + 0 \times 8 + 1 \times 4 + 0 \times 2 + 1 \times 1$	21_{10}
11011_2	$1 \times 16 + 1 \times 8 + 0 \times 4 + 1 \times 2 + 1 \times 1$	27_{10}
11100_2	$1 \times 16 + 1 \times 8 + 1 \times 4 + 0 \times 2 + 0 \times 1$	28_{10}
10001_2	$1 \times 16 + 0 \times 8 + 0 \times 4 + 0 \times 2 + 1 \times 1$	17_{10}
101010_2	$1 \times 32 + 0 \times 16 + 1 \times 8 + 0 \times 4 + 1 \times 2 + 0 \times 1$	42_{10}

3. Applying Binary Addition

Step 1: Add column by column from right to left.

Step 2: If sum = 2, write 0 and carry 1.

Step 3: If sum = 3, write 1 and carry 1.

Example 1	Example 2	Example 3
$\begin{array}{r} 111_2 \\ + 101_2 \\ \hline 1100_2 \end{array}$	$\begin{array}{r} 11101_2 \\ + 1101_2 \\ \hline 101010_2 \end{array}$	$\begin{array}{r} 1110_2 \\ + 111_2 \\ \hline 10101_2 \end{array}$

Step-by-Step:

Example 1: $111_2 + 101_2$

Right: $1+1=2 \rightarrow 0$ carry 1

Middle: $1+0+\text{carry}1=2 \rightarrow 0$ carry 1

Left: $1+1+\text{carry}1=3 \rightarrow 1$ carry 1

Extra: 1

Answer: 1100_2

Example 2: $11101_2 + 1101_2$

$1+1=2 \rightarrow 0$ carry 1

$0+0+\text{carry}1=1$

$1+1=2 \rightarrow 0$ carry 1

$1+1+\text{carry}1=3 \rightarrow 1$ carry 1

$1+0+\text{carry}1=2 \rightarrow 0$ carry 1

Extra: 1

Answer: 101010_2

Quick Practice:

A	B	A+B
101_2	10_2	111_2
1011_2	111_2	10010_2
1101_2	1110_2	11011_2
1111_2	1_2	10000_2
10101_2	111_2	11100_2
1111_2	1111_2	11110_2

4. Applying Binary Subtraction

Step 1: Subtract column by column from right to left.

Step 2: If 0-1, borrow 1 from the next column.

Step 3: Borrowed 1 = 2 in current column.

Example 1	Example 2	Example 3
$\begin{array}{r} 110_2 \\ - 1_2 \\ \hline 101_2 \end{array}$	$\begin{array}{r} 1010_2 \\ - 1_2 \\ \hline 1001_2 \end{array}$	$\begin{array}{r} 1101_2 \\ - 111_2 \\ \hline 1110_2 \end{array}$

Step-by-Step:

Example 1: $110_2 - 1_2$

Right: $0-1 \rightarrow$ borrow from middle

Middle becomes 0, right becomes 2

$2-1=1$

Middle: $0-0=0$

Left: $1-0=1$

Answer: 101_2

Example 2: $1010_2 - 1_2$

Right: $0-1 \rightarrow$ borrow from next

Next: 1 becomes 0, right becomes 2

$2-1=1$

Next: $0-0=0$

Next: $0-0=0$

Left: $1-0=1$

Answer: 1001_2

Quick Practice:

A	B	A-B
110_2	1_2	101_2
100_2	1_2	11_2
1010_2	10_2	1000_2
1100_2	11_2	1001_2
10001_2	111_2	1010_2
1111_2	101_2	1010_2

5. Applying Binary Addition with Multiple Numbers

Example 1	Example 2	Example 3
$\begin{array}{r} 111_2 \\ + 11_2 \\ + 1_2 \\ \hline 1011_2 \end{array}$	$\begin{array}{r} 11_2 \\ + 111_2 \\ + 1111_2 \\ \hline 11001_2 \end{array}$	$\begin{array}{r} 100111_2 \\ + 11_2 \\ + 1_2 \\ \hline 101011_2 \end{array}$

Quick Practice:

Problem	Answer
$111_2 + 11_2 + 1_2$	1011_2
$11_2 + 111_2 + 1111_2$	11001_2
$1011_2 + 111_2 + 1_2$	10011_2
$1110_2 + 110_2 + 1_2$	10101_2

6. Applying Binary Subtraction with Complex Numbers

Example 1	Example 2	Example 3
$\begin{array}{r} 10101_2 \\ - 11_2 \\ \hline 10010_2 \end{array}$	$\begin{array}{r} 10001_2 \\ - 101_2 \\ \hline 1100_2 \end{array}$	$\begin{array}{r} 1111_2 \\ - 111_2 \\ \hline 1000_2 \end{array}$

Quick Practice:

Problem	Answer
$1010_2 - 10_2$	1000_2
$1101_2 - 101_2$	1000_2
$10000_2 - 111_2$	1001_2
$1111_2 - 11_2$	1100_2

7. Application Checklist

■ BINARY APPLICATION CHECKLIST:

CONVERT DECIMAL → BINARY:

- Divide by 2 repeatedly
- Write quotient and remainder each time
- Stop when quotient = 0
- Read remainders from BOTTOM to TOP

CONVERT BINARY → DECIMAL:

- Identify place values (powers of 2)
- Multiply each digit by its place value
- Add all results together

BINARY ADDITION:

- Add from right to left (column by column)
- $0+0=0$, $1+0=1$, $1+1=10$ (carry 1)
- Continue carrying to next column

BINARY SUBTRACTION:

- Subtract from right to left (column by column)
- $0-1 \rightarrow$ borrow from next column
- Borrowed 1 = 2 in current column
- Continue borrowing as needed

CHECK YOUR ANSWER:

- Convert back to decimal to verify
- Addition: check if sum makes sense
- Subtraction: check if difference makes sense

8. Self-Test – Application Questions

No.	Question	Answer
1	Convert 19_{10} to binary	10011_2
2	Convert 31_{10} to binary	11111_2
3	Convert 1101_2 to decimal	13_{10}
4	Convert 10101_2 to decimal	21_{10}
5	Add: $101_2 + 11_2$	1000_2
6	Add: $1101_2 + 1010_2$	10111_2
7	Subtract: $110_2 - 1_2$	101_2
8	Subtract: $1000_2 - 11_2$	101_2
9	Add: $111_2 + 111_2 + 1_2$	1111_2
10	Subtract: $1111_2 - 101_2$	1010_2

Revision Tip: Practice applying conversion and arithmetic operations with different numbers. Start with simple numbers, then move to more complex ones. Always check your answers by converting back to decimal!